

Graph-Theoretic Approach to Controllability and Localizability of Decentralized Control Systems

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Abstract—The controllability and the localizability problems are considered under the decentralized information structure using some concepts from graph theory.

First of all, the information structure graph is introduced for a decentralized control system based upon local output controllability matrices between stations when a local output feedback law is applied, and some invariant properties of the graph with respect to local output feedback laws are driven.

Then by using these results, necessary and sufficient conditions of controllability under decentralized information structure are obtained. Also obtained are several conditions for localizability of decentralized control systems, which is newly introduced in the present paper as an extended concept of a class of disturbance localization and decoupling for centralized systems.

I. INTRODUCTION

RECENTLY, much attention has been paid to the decentralized control systems. To consider the control problems for decentralized control systems, it is very important to understand the information structure of the system clearly. Several studies have been done by using graphs to represent the essential information structure of the system. For example, Ho and Chu [1] used the precedence diagram to study dynamic team optimization problems, and Corfmat and Morse [2] defined a directed graph of the system based upon transfer matrices between stations to study the pole-assignability and stabilizability problems of decentralized control systems. Davison [3] introduced the concept of connectability to show the hierarchical structure of composite systems and studied the controllability and the observability under local output feedback gains in a usual sense.

In this paper, several basic properties of the directed graph of Corfmat and Morse, which will be called the information structure graph hereafter, will be made clear, and based upon this result the controllability and localiza-

bility problems of decentralized control systems are considered.

In Section II, the information structure graph of a linear decentralized system under a local output feedback law is defined as follows: it has an arc from station i to station j if the output controllability matrix from station i to station j is not equal to zero when the law is applied to the system. This means that the action of station i affects the observation of station j . The graphs for different feedback laws may not be the same, but it is shown that there is the maximum graph among the set of graphs for all possible local output feedback laws. This maximum graph is transitive and includes the others as its subgraphs. It is also shown that the existence of a path in the graphs is invariant with respect to local output feedback laws, i.e., if some graph has a path from station i to station j , then any other graph in the set also has the path.

Using these results, the controllability is studied in Section III. First of all, the quotient decentralized control system is introduced for the system to show the hierarchical structure clearly. Then necessary and sufficient conditions of the controllability under decentralized information structure are obtained. These are the natural extension of the results in [4] for systems with two stations to the general case of N stations.

In Section IV, a localizability problem is considered. This problem can be regarded as an extension of a class of disturbance localization and decoupling problems for centralized control systems [5]. The invariant property of paths with respect to local output feedback laws implies that it is impossible to separate station j from station i perfectly with local output feedback laws if there is a path from station i to station j in some information structure graph. But it may be possible to localize stations by a specific station i_0 , that is, to make the action of any station not affect the observation of other stations directly, but only through station i_0 . This localization will be useful in the sense that, by regarding station i_0 as a coordinator or higher level controller which does not apply any control to the system directly, the original system is decomposed into

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$(N-1)$ subsystems. Taking this fact into consideration, the localizability is defined as follows: the system is localizable by station i_0 if it is possible by an appropriate local output feedback law to make the action of any station not affect the observation of others directly, but through station i_0 . By introducing a distance of arcs to the information structure graph, several conditions for localizability are obtained.

II. INFORMATION STRUCTURE GRAPH

In this section, several basic properties of the directed graph which is introduced by Corfmat and Morse [2] are made clear. Although they defined it by using transfer matrices between two stations, it is defined in a slightly different way, i.e., by using the output controllability matrices.

Let us consider a decentralized control system with N stations (channels):

$$\dot{x}(t) = Ax(t) + B_1 u_1(t) + \cdots + B_N u_N(t), \quad (1a)$$

$$y_i(t) = C_i x(t), \quad (i = 1, 2, \dots, N \text{ and } t \in [0, \infty)) \quad (1b)$$

where $x(t)$ is an n -dimensional state vector, $u_i(t)$ and $y_i(t)$ are an r_i -dimensional control vector and an m_i -dimensional observation vector of station i , respectively.

It is assumed that every station knows matrices A, B_i, C_i ($i = 1, 2, \dots, N$) and the control laws of other stations, and the set of information available for station i at time t , $I_i(t)$, is given by

$$I_i(t) = \{y_i(p), u_i(q); p \in [0, t], q \in [0, t]\}. \quad (2)$$

Let D_i be a local output feedback gain of station i :

$$u_i(t) = D_i y_i(t) + v_i(t), \quad (3)$$

let $D = (D_1, D_2, \dots, D_N)$ be a local output feedback law, and let $\mathcal{D}$ be the set of all possible D . Then $\mathcal{D}$ can be regarded as an $\sum_{s=1}^N r_s \times m_s$ -dimensional Euclid space. For simplicity, the following notation is used:

$$A(D) = A + B_1 D_1 C_1 + \cdots + B_N D_N C_N. \quad (4)$$

Definition 1: The information structure graph $G(D)$ of system (1) under the local output feedback law D is defined to be the graph $\{N, \mathcal{Q}(D), \Delta(D)\}^1$ which has an arc a , i.e., $a \in \mathcal{Q}(D)$, such that $\Delta(D)(a) = (p, q)$, if and only if

$$C_q [B_p : A(D)B_p : \cdots : A(D)^{n-1}B_p] \neq 0. \quad (5)$$

Definition 1 means that if there is an arc $a \in \mathcal{Q}(D)$ such that $\Delta(D)(a) = (p, q)$; then the action of station p affects the observation of station q , when D is applied. Note that

¹Where N is a set of nodes which corresponds to a set of stations, $\mathcal{Q}(D)$ a set of arcs, and $\Delta(D)$ a directed incidence mapping, i.e., $\Delta(D)(a) = (p, q)$ means that $G(D)$ has an arc from p to q [6].

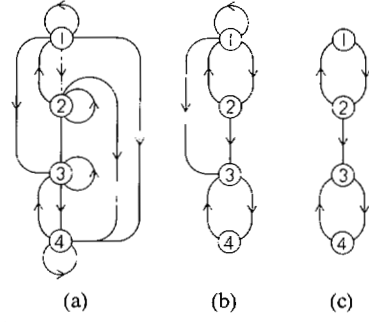


Fig. 1. (a) Directed graph $G(D)$ for $D_1 = (0, 0, 0, 0)$, (b) $D_2 = (-1, -1, 0, -1)$, and (c) $D_3 = (-1, -1, -1, -1)$.

the relation (5) is equivalent to

$$C_q [sI_n - A(D)]^{-1} B_p \neq 0. \quad (6)$$

Hence the information structure graph is the same as the directed graph introduced in [2]. Since if (5) is satisfied, station p can transmit any information to station q in a finite time interval (see Section III, Proposition 3), the graph can be regarded as representing the structure of the system with respect to the information transmission. This is the reason why the name of information structure graph is adopted.

Example 1: Consider the following system:

$$\begin{aligned} \dot{x}(t) = & \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_1(t) \\ & + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} u_2(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u_3(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u_4(t), \end{aligned} \quad (7a)$$

$$y_1(t) = [0 \ 1 \ 0 \ 0] x(t), \quad (7b)$$

$$y_2(t) = [1 \ 0 \ 0 \ 0] x(t), \quad (7c)$$

$$y_3(t) = [0 \ 1 \ 0 \ 1] x(t), \quad (7d)$$

$$y_4(t) = [0 \ 0 \ 1 \ 0] x(t). \quad (7e)$$

For this system, we consider three kinds of local output feedback laws as follows:

$$D_1 = (0, 0, 0, 0), \quad (7f)$$

$$D_2 = (-1, 0, -1, -1), \quad (7g)$$

$$D_3 = (-1, -1, -1, -1). \quad (7h)$$

Information structure graphs $G(D_i)$ ($i = 1, 2$, and 3) are different from one another as shown in Fig. 1.

However, the following proposition holds.

Proposition 1: The information structure graphs $G(D)$ are the same for almost all D , i.e., there exists a directed graph $G^\circ = \{N, \mathcal{Q}^\circ, \Delta^\circ\}$ such that

$$G^\circ = G(D), \quad \text{for almost all } D \in \mathcal{D}. \quad (8)$$

Moreover, $G(D)$ is a subgraph of G° for any $D \in \mathcal{D}$.

Proof: Consider the following polynomial:

$$f(\mathbf{D}; p, q) = \left\| C_q \begin{bmatrix} B_p & A(\mathbf{D})B_p & \cdots & A(\mathbf{D})^{n-1}B_p \end{bmatrix} \right\|^2 \quad (9)$$

where $\|M\|^2$ denotes the sum of squares of all the elements of a matrix M . Function $f(\mathbf{D}; p, q)$ is a finite-dimensional real coefficient polynomial with $\sum_{s=1}^N r_s \times m_s$ indeterminates. By Definition 1, if there is a $\mathbf{D}' \in \mathcal{D}$ for which the graph $G(\mathbf{D}')$ has an arc from p to q , then $f(\mathbf{D}'; p, q) \neq 0$, i.e., $f(\mathbf{D}; p, q)$ is not trivial for the pair (p, q) . Therefore, the existence of the arc from p to q is generic in $\mathcal{D}$ and this means that for almost all $\mathbf{D} \in \mathcal{D}$, $G(\mathbf{D})$ has an arc from p to q [5]. Since the number of nodes is finite, Proposition 1 holds. $\square$

Proposition 1 means that G° is the maximum one of $G(\mathbf{D})$. Hence we call G° the maximum information structure graph of system (1). The graph G° has the following property.

Proposition 2:

- 1) If there is a path from i to j in $G(\mathbf{D})$ for some $\mathbf{D}$, then there is an arc from i to j in G° .
- 2) Conversely, if there is an arc from i to j in G° , then there is a path from i to j in $G(\mathbf{D})$ for any $\mathbf{D} \in \mathcal{D}$.

Proof: See Appendix A. $\square$

Corollary 2.1: The existence of the path from i to j in the information structure graph is invariant with respect to local output feedback laws.

Corollary 2.2: The maximum information structure graph G° is the minimum transitive graph generated from $G(\mathbf{D})$ for any $\mathbf{D} \in \mathcal{D}$.

Proofs of Corollaries 2.1 and 2.2 are omitted because they are straightforward from Proposition 2.

The localization problem discussed in Section IV is motivated by Corollary 2.1. By Corollary 2.2, if an information structure graph is given, we can easily obtain the maximum information structure graph G° even without knowing the value of $\mathbf{D}$. For example, in Example 1, the minimum transitive graph of $G(\mathbf{D}_3)$ is $G(\mathbf{D}_1)$. Hence $G(\mathbf{D}_1)$ is the maximum information structure graph of system (7).

Using the graph G° , a pseudo-order² " $\rightarrow$ " and an equivalence relation " $\sim$ " can be defined among stations as follows.

Definition 2: The relations " $\rightarrow$ " and " $\sim$ " are defined by

- 1) $i \rightarrow i$ for any $i \in N$,
- 2) $i \rightarrow j$ if and only if there is an arc from i to j in G° ,
- 3) $i \sim j$ if and only if $i \rightarrow j$ and $j \rightarrow i$.

It is clear that the relation " $\rightarrow$ " is a pseudo-order and the relation " $\sim$ " is an equivalence relation.

The set of nodes, N , can be divided into several groups in such a way that any pair of nodes are equivalent if and only if they are in the same group. Then the restriction of

the graph to each of these groups makes a strongly connected graph. From Corollary 2.2, this division is independent of the local output feedback law $\mathbf{D}$.

III. CONTROLLABILITY UNDER DECENTRALIZED INFORMATION STRUCTURE

The controllability under decentralized information structure has been defined, and necessary and sufficient conditions are obtained for a system with two stations in [4]. In this section, this will be extended to the case of N stations based upon the results given in the previous section. The controllability is defined as follows [4].

Definition 3: System (1) is controllable under the decentralized information structure if there exist a finite number T and a decentralized control law such that $x(T) = 0$ for any unknown initial state vector $x(0)$.

Several preparations will be made for stating the main result, Proposition 6. The following proposition given in [4] is necessary in this section, too.

Proposition 3 [4, Proposition 1]: If $i \rightarrow j$, then station i can transmit necessary information about the initial state to station j in a finite time interval with signaling strategies, i.e., the unobservable subspace of station j is reduced to $\mathcal{H}_i \cap \mathcal{H}_j$, where $\mathcal{H}_i$ and $\mathcal{H}_j$ are the unobservable subspace of station i and station j , respectively.

Now assume that system (1) has $\tilde{N}$ ($\leq N$) equivalence classes with respect to the relation " $\sim$." Let N_i be a subset of N corresponding to the i th equivalence class.

Definition 4: The following system is called a quotient decentralized control system for system (1):³

$$\dot{x}(t) = Ax(t) + \tilde{B}_1 \tilde{u}_1(t) + \cdots + \tilde{B}_{\tilde{N}} \tilde{u}_{\tilde{N}}(t), \quad (10a)$$

$$\tilde{y}_i(t) = \tilde{C}_i x(t), \quad (10b)$$

where

$$i = 1, 2, \dots, \tilde{N},$$

$$\tilde{B}_i = [B_{i_1} \ B_{i_2} \ \cdots \ B_{i_r}], \quad (10c)$$

$$\tilde{C}_i = \begin{bmatrix} C_{i_1} \\ C_{i_2} \\ \vdots \\ C_{i_r} \end{bmatrix}, \quad (10d)$$

and

$$N_i = \{i_1, i_2, \dots, i_r\}. \quad (10e)$$

Note that $\tilde{B}_i(\tilde{C}_i)$ is composed of the driving (observation) matrices of stations belonging to the i th equivalence class. Let $\tilde{\mathcal{H}}_i$ be the controllable subspace and let $\tilde{\mathcal{H}}_i$ be the unobservable subspace for the triple $(\tilde{C}_i, A, \tilde{B}_i)$, i.e.,

²A relation R on a set is called a pseudo-order, if aRa for all elements and aRb and bRc means aRc for any triple a, b , and c of the set.

³The terminology "quotient decentralized control system" is adopted from "quotient set" in algebraic theory.

$$\begin{aligned}\tilde{\mathcal{B}}_i &= \text{Im}\{\tilde{B}_i\}, \\ \tilde{\mathcal{N}}_i &= \text{Ker}\{\tilde{C}_i\}, \\ \tilde{\mathcal{R}}_i &= \langle A|\tilde{\mathcal{B}}_i \rangle \\ &= \tilde{\mathcal{B}}_i + A\tilde{\mathcal{B}}_i + A^2\tilde{\mathcal{B}}_i + \cdots + A^{n-1}\tilde{\mathcal{B}}_i, \quad (11)\end{aligned}$$

$$\tilde{\mathcal{K}}_i = \bigcap_{s=1}^N A^{s-1}\tilde{\mathcal{N}}_i \quad (12)$$

where $\text{Im}\{M\}$ and $\text{Ker}\{M\}$ are the range and the null space of a matrix M , respectively. By using these subspaces, we define the subspace $\tilde{\mathcal{X}}_i$ as follows:

$$\tilde{\mathcal{X}}_i = \tilde{\mathcal{R}}_i \cap \tilde{\mathcal{K}}_i^\perp, \quad (13)$$

where $\mathcal{K}^\perp$ denotes the orthocomplement of $\mathcal{K}$ in the n -dimensional vector space. $\tilde{\mathcal{X}}_i$ has the following properties.

Proposition 4: The subspace $\tilde{\mathcal{X}}_i$ is invariant with respect to local output feedback law $\tilde{D}$.

The proof of this proposition is given in Appendix B. The following proposition is given as a simple extension of Proposition 3.

Proposition 5: If

$$x(0) \in \tilde{\mathcal{X}}_i, \quad (14)$$

then stations of the i th equivalence class can move $x(0)$ to the origin in a finite time interval in cooperation with one another.

Proof: By Proposition 3, all of the stations of the i th equivalence class can identify $x(0)$ in a finite time interval, say $[0, T_1]$, by exchanging information by using, for example, the signaling strategies in [4] or the periodic time-varying output feedback law in [7]. After this, they are to reconstruct $x(T_1)$ and control it into the origin in cooperation with one another in a finite time interval. This is possible because $x(T_1) \in \tilde{\mathcal{R}}_i$ for any action of the stations of the i th class. $\square$

Propositions 4 and 5 mean that the subspace $\tilde{\mathcal{X}}_i$ is the invariant observable and controllable subspace of the i th equivalence class. For the quotient system (10), we define a linear order relation as follows:

1) $a \Rightarrow b$ if and only if not $b \rightarrow a$,

2) $a \Rightarrow b$ or $b \Rightarrow a$ for each pair a and b ,

where a and b are elements of $\tilde{N} = \{1, 2, \dots, \tilde{N}\}$. This is possible because the information structure graph of system (10) has no loop except for trivial ones, i.e., arcs from a node to itself. Now assume without loss of generality that

$$1 \Rightarrow 2 \Rightarrow \cdots \Rightarrow \tilde{N}. \quad (15)$$

Let $\tilde{F}_i$ be the Hankel matrix of the triple $(\tilde{C}_i, A, \tilde{B}_i)$, i.e.,

$$\tilde{F}_i = \begin{bmatrix} \tilde{C}_i \tilde{B}_i & \cdots & \tilde{C}_i A^{n-1} \tilde{B}_i \\ \vdots & & \vdots \\ \tilde{C}_i A^{n-1} \tilde{B}_i & \cdots & \tilde{C}_i A^{2n-2} \tilde{B}_i \end{bmatrix} \quad (16)$$

and let $\tilde{\Lambda}$ be the set of fixed modes of the quotient system (10)

$$\tilde{\Lambda} = \bigcap_{\tilde{D} \in \tilde{\mathcal{D}}} \left\{ s \in \mathcal{C} : \det \left[sI_n - \left(A + \sum_{i=1}^{\tilde{N}} \tilde{B}_i \tilde{D}_i \tilde{C}_i \right) \right] = 0 \right\} \quad (17)$$

where $\mathcal{C}$ is the complex number field, $\tilde{D}_i$ a local output feedback gain of the i th equivalence class, $\tilde{D}$ a local output feedback law, and $\tilde{\mathcal{D}}$ a set of all possible $\tilde{D}$ [4]. With these preparations we can now state the following main result on controllability.

Proposition 6: The following conditions are equivalent.

1) System (1) is controllable under the decentralized information structure.

2) $\sum_{i=1}^N \text{rank } \tilde{F}_i = n$.

3) The quotient system of (1) has no fixed mode, i.e., $\tilde{\Lambda} = \emptyset$.

Proof: See Appendix C. $\square$

Conditions 2) and 3) of Proposition 6 are the extensions of Theorem 1 and Theorem 2 in [4] to the general case of N stations.

Example 2: For system (7) of Example 1, we have $1 \sim 2$, $3 \sim 4$, and $2 \rightarrow 3$. Hence the quotient system of system (7) is obtained as follows:

$$\begin{aligned}\dot{x}(t) &= \begin{bmatrix} 2 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} \tilde{u}_1(t) \\ &\quad + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} \tilde{u}_2(t), \quad (18a)\end{aligned}$$

$$\tilde{y}_1(t) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x(t), \quad (18b)$$

$$\tilde{y}_2(t) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x(t). \quad (18c)$$

For $\tilde{D} = (\tilde{D}_1, \tilde{D}_2)$,

$$\begin{aligned}\det[sI_4 - (A + \tilde{B}_1 \tilde{D}_1 \tilde{C}_1 + \tilde{B}_2 \tilde{D}_2 \tilde{C}_2)] \\ = \det \left[sI_2 - \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} - \tilde{D}_1 \right] \\ \cdot \det \left[sI_2 - \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix} - \tilde{D}_2 \right]. \quad (19)\end{aligned}$$

Since $\tilde{D}_1$ and $\tilde{D}_2$ are arbitrary matrices, we have

$$\tilde{\Lambda} = \emptyset. \quad (20)$$

Hence by Proposition 6, system (7) is controllable under the decentralized information structure.

Example 3: Consider a system described by

$$\begin{aligned} \dot{x}(t) = & \begin{bmatrix} 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_1(t) \\ & + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \\ 0 \end{bmatrix} u_2(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u_3(t) \\ & + \begin{bmatrix} 0 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_4(t) + \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} u_5(t), \quad (21a) \\ y_1(t) = & [0 \ 1 \ 0 \ 0 \ 0] x(t), \quad (21b) \\ y_2(t) = & [0 \ 1 \ 0 \ 0 \ 0] x(t), \quad (21c) \\ y_3(t) = & [0 \ 0 \ 1 \ 0 \ 1] x(t), \quad (21d) \\ y_4(t) = & [1 \ 0 \ 0 \ 0 \ 0] x(t), \quad (21e) \\ y_5(t) = & [0 \ 0 \ 0 \ 1 \ 0] x(t). \quad (21f) \end{aligned}$$

Fig. 2 shows the graph $G((0,0,0,0,0))$ for system (21). From the figure, we see that the system has three equivalence classes, $\{1,4\}$, $\{2\}$, $\{3,5\}$, and

$$\{1,4\} \Rightarrow \{2\} \Rightarrow \{3,5\}. \quad (22)$$

It is easy to see that

$$\tilde{\Lambda} = \{0\} \neq \emptyset. \quad (23)$$

Hence this system is not controllable under the decentralized information structure. In fact, if

$$x(0) = (0,0,1,0,0)^t \quad (24)$$

where x^t is the transposed vector of x , then stations 3 and 5 can know the initial state $x(0)$, hence, of course, the third component of $x(0)$. But they cannot control it. On the other hand, station 2, the only station that can control the third component of the state, cannot distinguish the state from the zero state.

IV. LOCALIZATION PROBLEM

A new concept of localizability of a decentralized control system is introduced and several conditions for localizability are given in this section. These are obtained by using the information structure graph along with a concept of distance between its nodes. First of all, the localizability of system (1) is defined as follows.

Definition 4: Stations $1, 2, \dots, N-1$ of system (1) is said to be localizable by station N , if there is a local output feedback law D such that

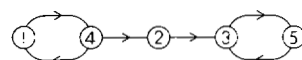


Fig. 2. Information structure graph of system (21).

$$C_j [B_i : A(D)B_i : \dots : A(D)^{n-1}B_i] = 0, \quad (25)$$

for $i, j = 1, 2, \dots, N-1$ and $i \neq j$, or equivalently

$$\mathcal{N}_i^* \supset \langle A(D) | \mathcal{B}_i \rangle \quad (26)$$

where $i = 1, 2, \dots, N-1$,

$$\begin{aligned} \mathcal{N}_i &= \text{Ker}\{C_i\}, \\ \mathcal{N}_i^* &= \bigcap_{j=1}^{N-1} \mathcal{N}_j, \\ \mathcal{B}_i &= \text{Im}\{B_i\}. \end{aligned} \quad (27)$$

Definition 4 means that if stations $1, 2, \dots, N-1$ of system (1) is localized by station N , then the input of station i does not directly affect the observation of station j ($i, j = 1, 2, \dots, N-1$ and $i \neq j$). We may be able to consider station N as a higher level coordinator of system (1). Note that Definition 4 does not deal with the complete localization of stations where the input of any station does not affect the observation of the other stations, since Corollary 2.1 implies that such localization is impossible except for very trivial systems.

The above defined localization can be regarded as an extended concept of decoupling and disturbance localization for centralized systems. For example, a disturbance localization problem by state feedback for centralized systems is formulated as follows [5]. For system

$$\dot{x}(t) = Ax(t) + Bu(t) + Ew(t), \quad (28a)$$

$$y(t) = Cx(t), \quad (28b)$$

find a matrix F such that

$$\text{Ker}\{C\} \supset \langle A + BF | \text{Im}\{E\} \rangle.$$

Now consider a decentralized system:

$$\dot{x}(t) = Ax(t) + B_1u_1(t) + B_2u_2(t) + B_3u_3(t), \quad (29a)$$

$$B_1 = E, \quad B_2 = B, \quad B_3 = B,$$

$$u_1(t) = w(t),$$

$$u_2(t) = u(t) - v(t),$$

$$u_3(t) = v(t),$$

$$y_1(t) = 0, \quad (29b)$$

$$y_2(t) = Cx(t), \quad (29c)$$

$$y_3(t) = x(t). \quad (29d)$$

Then the above disturbance localization problem can be shown to be equivalent to the localization problem of this system by station 3, which performs the necessary state feedback.

Consider a decoupling of the following centralized system by state feedback:

$$\dot{x}(t) = Ax(t) + [B_1 \ B_2 \ \cdots \ B_{N-1}] \begin{bmatrix} u_1(t) \\ u_2(t) \\ \vdots \\ u_{N-1}(t) \end{bmatrix}, \quad (30a)$$

$$y(t) = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_{N-1}(t) \end{bmatrix} = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_{N-1} \end{bmatrix} x(t). \quad (30b)$$

We assume that the desirable pairing of input and output variables is given beforehand, i.e., it is desired to make the input $u_i(t)$ not affect the output $y_j(t)$ ($j \neq i$; $i, j = 1, 2, \dots, N-1$). Hence, the problem is to find a matrix F such that

$$\bigcap_{\substack{j=1 \\ j \neq i}}^{N-1} \text{Ker}\langle C_j \rangle \supset \langle A + BF[\text{Im}\langle B_i \rangle] \rangle$$

for $i = 1, 2, \dots, N-1$. If we consider a decentralized system

$$\begin{aligned} \dot{x}(t) &= Ax(t) + \sum_{i=1}^{N-1} B_i u_i(t) \\ &\quad + [B_1 \ B_2 \ \cdots \ B_{N-1}] u_N(t), \\ y_i(t) &= C_i x(t), \quad i = 1, 2, \dots, N-1 \\ y_N(t) &= x(t), \end{aligned}$$

then the above decoupling problem is equivalent to the localization problem of this system by station N .

It is now seen that disturbance localization and decoupling problems for centralized systems can be reduced to equivalent localization problems for decentralized systems, at least conceptually. Also note that Definition 4 is a natural extension of output decomposition problems of decentralized linear multivariable systems by output feedback treated in [8] to systems with more than two stations.

We assume that $N \geq 3$ hereafter, since Definition 4 is meaningless for $N \geq 2$.

Now as a preparation for considering the localization problem, a distance between nodes of the information structure graph is defined and some properties of the distance are discussed.

Definition 5: For $G(\mathbf{D}) = \{N, \mathcal{Q}(\mathbf{D}), \Delta(\mathbf{D})\}$ of system (1), the length of the arc $a \in \mathcal{Q}(\mathbf{D})$ from i to j is the least positive integer α such that

$$C_j \left(A + \sum_{s=1}^N B_s D_s C_s \right)^{\alpha-1} B_i \neq 0. \quad (31)$$

The length of a path p from i to j , $\rho(p)$, is the sum of the

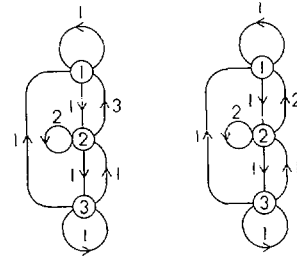


Fig. 3. Length of arcs of system (3).

lengths of all arcs included in p . The distance from i to j in $G(\mathbf{D})$ is defined as follows:

$$\rho(i, j; \mathbf{D}) = \min_p \rho(p) \quad (32)$$

where the minimum operation is taken over all possible paths from i to j in $G(\mathbf{D})$.

Example 4: Consider a system:

$$\begin{aligned} \dot{x}(t) &= \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} u_1(t) \\ &\quad + \begin{bmatrix} 0 \\ 0 \\ 1 \\ 0 \end{bmatrix} u_2(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix} u_3(t), \end{aligned} \quad (33a)$$

$$y_1(t) = [1 \ 0 \ 0 \ 0] x(t), \quad (33b)$$

$$y_2(t) = [1 \ 1 \ 0 \ 1] x(t), \quad (33c)$$

$$y_3(t) = [0 \ 0 \ 1 \ 1] x(t). \quad (33d)$$

Graphs $G((0,0,0))$ and $G((0,0,a))$ with $a \neq 0$ are given in Fig. 3 along with the length of arcs. While the length of the arc from 2 to 1 is 3 in $G((0,0,0))$, it is 2 in the other. But in both graphs, the distance from 2 to 1 is the same, i.e.,

$$\rho(2, 1; (0,0,0)) = \rho(2, 1; (0,0,a)) = 2. \quad (34)$$

Based upon Definition 5 we will show the existence of $\mathbf{D}^*$ for which the distance between any pair of nodes in $G(\mathbf{D}^*)$ is minimal.

Proposition 7: There is a graph $G(\mathbf{D}^*)$ such that

1) if there is a path from i to j in some $G(\mathbf{D})$, then there also exists the same path in $G(\mathbf{D}^*)$,

$$2) \rho(i, j; \mathbf{D}^*) \leq \rho(i, j; \mathbf{D}) \text{ for all } \mathbf{D}, \text{ and} \quad (35)$$

$$3) G(\mathbf{D}^*) = G(\mathbf{D}) \text{ for almost all } \mathbf{D} \in \mathcal{D}. \quad (36)$$

Proof: Assume that some $G(\mathbf{D})$ has an arc from p to q and define a polynomial $g(\mathbf{D}, \alpha; p, q)$ and a subset of positive integers $P(p, q)$ as follows:

$$g(\mathbf{D}, \alpha; p, q) = \|C_q A(\mathbf{D})^{\alpha-1} B_p\|^2, \quad (37)$$

$$P(p, q) = \{\alpha: \text{there is a } \mathbf{D} \in \mathcal{D} \text{ such that}$$

$$g(\mathbf{D}, \alpha; p, q) \neq 0\}. \quad (38)$$

Then $P(i, j) \neq \emptyset$ by the assumption. Hence $P(i, j)$ has the minimum element α^* . Therefore there is a D' such that $g(D; \alpha^*; i, j) \neq 0$. This means that $g(D, \alpha^*; i, j) \neq 0$ for almost all D . Let $\mathcal{D}(i, j)$ be a subset of $\mathcal{D}$ such that

$$g(D, \alpha^*; i, j) \neq 0 \quad \text{for all } D \in \mathcal{D}(i, j); \quad (39)$$

then $\mathcal{D}(i, j)$ is generic in $\mathcal{D}$. Let us define $\mathcal{D}^*$ as follows:

$$\mathcal{D}^* = \bigcap_{(i, j) \in \mathcal{A}} \mathcal{D}(i, j); \quad (40)$$

then $\mathcal{D}^*$ is also generic in $\mathcal{D}$ since the set of arcs, $\mathcal{A}$, is a finite set. It is obvious that $G(D^*)$ ($D^* \in \mathcal{D}^*$) has the properties 1), 2), and 3). $\square$

Because of (35), we call $\rho(i, j; D^*)$ the distance from i to j and define it by $\rho(i, j)$. We also express $G(D^*)$ by $G^* (= \{N, \mathcal{A}, \Delta, \rho\})$. It is obvious that G^* is equal to G° in Section II except for the distance. G^* has the properties given in the following two propositions.

Proposition 8: For G^* , the distance from i to j is equal to the length of the arc from i to j .

Proof: Assume that $G^* = G(\mathbf{0})$ where $\mathbf{0} = (0, 0, \dots, 0)$ and that

$$a_1, a_2 \in \mathcal{A}, \quad (41)$$

$$\Delta(a_1) = (1, 2), \quad (42)$$

$$\Delta(a_2) = (2, 3), \quad (43)$$

and let γ and δ be the length of a_1 and a_2 , respectively. Then by assumption, we have

$$C_2 A^{\gamma-1} B_1 \neq 0, \quad (44)$$

$$C_3 A^{\delta-1} B_2 \neq 0. \quad (45)$$

Therefore, we have

$$\begin{aligned} C_3 (A + B_2 D_2 C_2)^{\gamma+\delta-1} B_1 \\ = C_3 A^{\gamma+\delta-1} B_1 + C_3 A^{\delta-1} B_2 D_2 C_2 A^{\gamma-1} B_1. \end{aligned} \quad (46)$$

It is easy to find a matrix D_2 for which (46) is not equal to zero. Hence there is an arc from 1 to 3 in G^* and the distance is less than or equal to $\gamma + \delta$.

This means that the distance satisfies the triangular inequality and Proposition 8 is true. $\square$

Proposition 9: If there is an arc from i to j in G^* , then there is a path from i to j in $G(D)$ and

$$\rho(i, j; D) = \rho(i, j) \quad (47)$$

for any $D \in \mathcal{D}$.

Proof: The former statement is trivial since G^* is equal to G° except for the distance. For the latter statement, we will show that if there is a path from 1 to 2 in $G(\mathbf{0})$ (where $G(\mathbf{0})$ may or may not be equal to G^*), then

$$\rho(1, 2; \mathbf{0}) = \rho(1, 2). \quad (48)$$

Now assume that $D \in \mathcal{D}^*$ [see (40)] and that $\alpha = \rho(1, 2)$; then we have

$$C_2 A(D)^{\alpha-1} B_1 \neq 0 \quad (49)$$

by Proposition 8. If

$$C_2 A^{\alpha-1} B_1 = 0, \quad (50)$$

then there is an arc from 1 to 2 in $G(\mathbf{0})$ and the length is less than or equal to α . By the property of G^* , (48) holds. When (50) is not true, there are positive numbers $s_1, s_2, \dots, s_{r-1}, p_1, p_2, \dots, p_{r-1}$ and p_r such that

$$C_2 A^{p_r-1} B_{s_{r-1}} D_{s_{r-1}} C_{s_{r-1}} \cdots C_{s_1} A^{p_1-1} B_1 \neq 0. \quad (51)$$

This implies that there is a set of arcs a_i ($i = 1, 2, \dots, r$) in $G(\mathbf{0})$ such that

$$\Delta(a_1) = (1, s_1), \quad (52)$$

$$\Delta(a_i) = (s_{i-1}, s_i), \quad i = 2, 3, \dots, r-1, \quad (53)$$

$$\Delta(a_r) = (s_r, 2). \quad (54)$$

These arcs compose a path from 1 to 2 in $G(\mathbf{0})$ and the length is $p_1 + p_2 + \dots + p_r = \alpha$. Hence the distance from 1 to 2 in $G(\mathbf{0})$ is less than or equal to α . Equation (48) holds again. $\square$

Proposition 9 asserts that the distance of the information structure graph is invariant with respect to local output feedback laws. Therefore we can define a center node (station) of the information structure graph.

Definition 6: We call the node i° a center node when the following property is satisfied: if there is an arc from i to j in G^* , then there also exist arcs from i to i° and from i° to j such that

$$\rho(i, j) = \rho(i, i^\circ) + \rho(i^\circ, j) \quad (55)$$

for $i, j = 1, 2, \dots, N$ and $i \neq j \neq i^\circ \neq i$.

By (55), we have

$$1 \leq \rho(i, i^\circ) < \rho(i, j), \quad (56)$$

$$1 \leq \rho(i^\circ, j) < \rho(i, j), \quad (57)$$

for $i, j = 1, 2, \dots, N$ and $i \neq j \neq i^\circ \neq i$. Equations (56) and (57) mean that a center node is the nearest node from any other nodes. With these preparations we now state several results on localizability.

Proposition 10: If stations 1, 2, $\dots$, and $N-1$ are localizable by station N , then the node N is a center node.

The proof of Proposition 10 is trivial from Proposition 9 and Definition 6 and is omitted here. Along with Proposition 10, the following proposition makes it very easy to consider the localization problems.

Proposition 11: If stations 1, 2, $\dots$, and $N-1$ are localizable by station N , then there exists a local feedback law $D^\circ = (0, 0, \dots, 0, D_N^\circ)$ such that

$$C_i A(D^\circ)^{p-1} B_j = 0 \quad (58)$$

for $i, j = 1, 2, \dots, N-1$ ($i \neq j$) and $p = 1, 2, 3, \dots$.

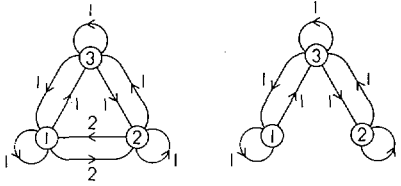


Fig. 4. Localization of stations 1 and 2 by station 3.

Proof: Assume that, for $\mathbf{D}^* = (D_1^*, D_2^*, \dots, D_N^*)$, (25) holds. Then for any $\mathbf{D} = (D_1, D_2, \dots, D_{N-1}, 0)$, we have

$$\begin{aligned} C_i A (\mathbf{D}^* + \mathbf{D})^{p-1} B_j \\ = C_i \left(A (\mathbf{D}^*) + \sum_{s=1}^{N-1} B_s D_s C_s \right)^{p-1} B_j = 0 \end{aligned} \quad (59)$$

for $i, j = 1, 2, \dots, N-1$ ($i \neq j$) and $p = 1, 2, 3, \dots$.

Therefore by setting D_i equal to $-D_i^*$ ($i = 1, 2, \dots, N-1$), Proposition 11 is proved. $\square$

Example 5: Consider the localization problem for the following system:

$$\begin{aligned} \dot{x}(t) = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_1(t) \\ + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_2(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u_3(t), \end{aligned} \quad (60a)$$

$$y_1(t) = [1 \ 0] x(t), \quad (60b)$$

$$y_2(t) = [0 \ 1] x(t), \quad (60c)$$

$$y_3(t) = [1 \ 1] x(t). \quad (60d)$$

The graph $G^* = G((0,0,0))$ is given in Fig. 4(a). Neither node 1 nor 2 is a center node because

$$\begin{aligned} \rho(3,1) + \rho(1,2) = 3 \\ \neq \rho(3,2) = 1 \end{aligned} \quad (61)$$

and

$$\begin{aligned} \rho(3,2) + \rho(2,1) = 3 \\ \neq \rho(3,1) = 1. \end{aligned} \quad (62)$$

But since

$$\rho(1,3) + \rho(3,2) = \rho(1,2) = 2, \quad (63)$$

$$\rho(2,3) + \rho(3,1) = \rho(2,1) = 2, \quad (64)$$

node 3 is a center node. Therefore, by Propositions 10 and 11, it may be possible to localize stations 1 and 2 by station 3 using a local feedback law $\mathbf{D}^0 = (0, 0, D_N^0)$. In fact, we can easily verify that they are localized by station 3 by using $\mathbf{D} = (0, 0, -1)$ as shown in Fig. 4(b). From this graph, it can be seen that the action of station 1 does not affect the observation of station 2 directly and vice versa.

It is easy to show that the sufficiency of Proposition 10 does not hold. For example, assume that we have

$$\begin{aligned} \dot{x}(t) = \begin{bmatrix} 2 & 2 \\ 1 & 2 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u_1(t) \\ + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u_2(t) + \begin{bmatrix} 1 \\ 1 \end{bmatrix} u_3(t) \end{aligned} \quad (65)$$

instead of (60a). Then the graph G^* is the same as that for system (60) and node 3 is a center node, but stations 1 and 2 are not localizable by station 3.

Now assume that

$$\text{rank} \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_{N-1} \end{bmatrix} = n \quad (66)$$

and

$$C_N = I_n. \quad (67)$$

Equation (66) means that

$$\bigcap_{i=1}^{N-1} \mathcal{U}_i = \{0\}. \quad (68)$$

Conditions (66) and (67) are called compatibility conditions [5].

Let $\mathcal{V}_i^*$ be the largest (A, B_N) -invariant subspace contained in $\mathcal{U}_i^*$, i.e.,

$$\mathcal{V}_i^* = \sup \{ \mathcal{V}_i : \mathcal{V}_i \subset \mathcal{U}_i^* \cap A^{-1}(\mathcal{B}_N + \mathcal{V}_i) \} \quad (69)$$

where $\mathcal{U}_i^*$ and $\mathcal{B}_N$ are defined by (27).

Then we have the following proposition as an extension of Theorem 4.2. in [5].

Proposition 12: Assume that (66) and (67) hold; then stations $1, 2, \dots$, and $N-1$ are localizable by station N if and only if

$$\mathcal{V}_i^* \supset \mathcal{B}_i, \quad i = 1, 2, \dots, N-1. \quad (70)$$

Proof:

Necessity: Assume that stations $1, 2, \dots$, and $N-1$ are localizable by station N ; then Proposition 11 and (67) imply that there is a matrix D_N such that

$$\langle A + B_N D_N | \mathcal{B}_i \rangle \subset \mathcal{U}_i^*. \quad (71)$$

Define

$$\mathcal{W}_i = \langle A + B_N D_N | \mathcal{B}_i \rangle; \quad (72)$$

then

$$\mathcal{W}_i \subset \mathcal{U}_i^* \cap A^{-1}(\mathcal{B}_N + \mathcal{W}_i). \quad (73)$$

Therefore, we have

$$\mathcal{B}_i \subset \mathcal{W}_i \subset \mathcal{V}_i^*. \quad (74)$$

Sufficiency: The relation

$$A\mathcal{V}_i^* \subset \mathcal{B}_N + \mathcal{V}_i^* \quad (75)$$

means that there is a map (or a matrix) F_i such that

$$(A + B_N F_i)\mathcal{V}_i^* \subset \mathcal{V}_i^*, \quad i = 1, 2, \dots, N-1. \quad (76)$$

By (68), we have

$$\mathcal{U}_i^* \cap \sum_{\substack{j=1 \\ j \neq i}}^{N-1} \mathcal{U}_j^* = \bigcap_{j=1}^{N-1} \mathcal{U}_j = \{0\} \quad (77)$$

and

$$\mathcal{V}_i^* \subset \mathcal{U}_i^*, \quad i = 1, 2, \dots, N-1. \quad (78)$$

Equations (77) and (78) mean that subspaces $\mathcal{V}_i^*$ ($i = 1, 2, \dots, N-1$) are linearly independent from one another. Hence there is a map (or a matrix) F such that

$$F|_{\mathcal{V}_i^*} = F_i, \quad i = 1, 2, \dots, N-1 \quad (79)$$

where $F|_{\mathcal{V}_i^*}$ denotes the restricted map of F in $\mathcal{V}_i^*$. Then we have

$$(A + B_N F)\mathcal{V}_i^* = (A + B_N F_i)\mathcal{V}_i^* \subset \mathcal{V}_i^* \subset \mathcal{U}_i^*; \quad (80)$$

then by (70), we have

$$\langle A + B_N F|_{\mathcal{B}_i} \rangle \subset \mathcal{U}_i^*. \quad (81)$$

Hence the stations are localized.

It is trivial that condition (67) can be relaxed as follows:

$$\mathcal{U}_N \subset \text{Ker}\{F\} \quad (82)$$

where F is given in (79). Condition (68) may also be relaxed if station N can use a dynamic compensator similar to [5, Theorem 9.3], but this means that stations $1, 2, \dots$, and $N-1$ transmit the value of their control vector to station N .

If station N is a center station of system (1), then it follows from (56) and (57) that

$$\mathcal{B}_i \subset \mathcal{U}_i^*, \quad i = 1, 2, \dots, N-1. \quad (83)$$

Therefore, the following corollary is easily given by Proposition 12.

Corollary 12.1: Assume that (66) and (67) hold and that

$$A\mathcal{B}_i \subset \mathcal{B}_i + \mathcal{B}_N \quad \text{or} \quad A\mathcal{U}_i^* \subset \mathcal{U}_i^* + \mathcal{B}_N, \quad i = 1, 2, \dots, N-1; \quad (84)$$

then stations $1, 2, \dots, N-1$ are localizable by station N if and only if station N is a center station of system (1).

Example 6: Consider the following system:

$$\dot{x}(t) = Ax(t) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \\ 0 & 0 \end{bmatrix} u_1(t) + \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u_2(t) + \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{bmatrix} u_3(t), \quad (85a)$$

$$y_1(t) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} x(t), \quad (85b)$$

$$y_2(t) = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} x(t), \quad (85c)$$

$$y_3(t) = x(t), \quad (85d)$$

and A is any n -dimensional system matrix.

For this system, station 3 is a center node and

$$A\mathcal{B}_i \subset \mathcal{B}_i + \mathcal{B}_3, \quad i = 1, 2 \text{ for any } A. \quad (86)$$

Hence stations 1 and 2 of system (85) are localizable by station 3 by Corollary 12.1, and the reduced subsystems are output controllable by stations 1 and 2, respectively.

Example 7: The following system satisfies (67) and (68) for $N = 4$, but stations 1, 2, and 3 are not localizable by station 4:

$$\dot{x}(t) = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{bmatrix} x(t) + \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} u_1(t) + \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} u_2(t) + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u_3(t) + \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix} u_4(t), \quad (87a)$$

$$y_1(t) = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix} x(t), \quad (87b)$$

$$y_2(t) = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix} x(t), \quad (87c)$$

$$y_3(t) = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix} x(t), \quad (87d)$$

$$y_4(t) = x(t), \quad (87e)$$

because station 4 is not a center node of the information structure graph of system (87).

V. CONCLUSIONS

In this paper, the information structure graph was defined for a decentralized control system by using output controllability matrices between stations, and it was shown that the existence of the path and the distance between stations are invariant under local output feedback laws.

By using these properties, necessary and sufficient conditions of controllability under decentralized information structure were obtained. Also obtained were several conditions for localizability of decentralized control systems, which was newly introduced in the present paper as an extended concept of a class of disturbance localization and decoupling for centralized systems.

APPENDIX A

Proof of Proposition 2:

1) Proposition 1 asserts that if there is a path from i to j in some $G(\mathbf{D})$, the same path exists in G° too. Therefore, it is enough to prove that G° is a transitive graph. We will show it by contradiction. Suppose that for $G^\circ = \{N, \mathcal{Q}, \Delta\}$,

$$1, 2, 3 \in N, \quad (\text{A1})$$

$$\Delta(a) = (1, 2), \quad (\text{A2})$$

$$\Delta(b) = (2, 3), \quad (\text{A3})$$

but that there does not exist any arc from 1 to 3 in G° . From (A2), (A3), and Definition 1, we have

$$C_2[B_1 : AB_1 : \cdots : A^{n-1}B_1] \neq 0, \quad (\text{A4})$$

$$C_3[B_2 : AB_2 : \cdots : A^{n-1}B_2] \neq 0, \quad (\text{A5})$$

$$C_3[B_1 : AB_1 : \cdots : A^{n-1}B_1] = 0, \quad (\text{A6})$$

where we assumed that $G^\circ = G((0, 0, \dots, 0))$ without loss of generality. Let δ and γ be the least positive integers such that

$$C_2 A^{\delta-1} B_1 \neq 0, \quad (\text{A7})$$

$$C_3 A^{\gamma-1} B_2 \neq 0, \quad (\text{A8})$$

respectively. Then, from (A6), we have

$$C_3(A + B_2 D_2 C_2)^{\delta+\gamma-1} B_1 = C_3 A^{\gamma-1} B_2 D_2 C_2 A^{\delta-1} B_1. \quad (\text{A9})$$

It is easy to find a matrix D_2 for which (A9) is not equal to zero. This means that there exists a $\mathbf{D}$ whose graph $G(\mathbf{D})$ has an arc from 1 to 3. But this contradicts the definition of G° , completing the proof of 1).

2) It will be shown that if there is not any path from i to j in $G(\mathbf{D}')$ for some $\mathbf{D}' \in \mathcal{D}$, then there is not an arc from i to j in G° . Let us define two subsets of N :

$$N'_1 = \{p \in N: p = i \text{ or there is a path from } i \text{ to } p \text{ in } G(\mathbf{D}')\}, \quad (\text{A10})$$

$$N'_2 = N - N'_1. \quad (\text{A11})$$

We assume that $i = 1, j = N$,

$$N'_1 = \{1, 2, \dots, r\}, \quad (\text{A12})$$

and

$$N'_2 = \{r+1, r+2, \dots, N\}, \quad (\text{A13})$$

without loss of generality. The following matrices are constructed according to sets (A12) and (A13):

$$B'_1 = [B_1 : B_2 : \cdots : B_r], \quad (\text{A14})$$

$$B'_2 = [B_{r+1} : B_{r+2} : \cdots : B_N], \quad (\text{A15})$$

$$C'_1 = \begin{bmatrix} C_1 \\ C_2 \\ \vdots \\ C_r \end{bmatrix}, \quad \text{and} \quad C'_2 = \begin{bmatrix} C_{r+1} \\ C_{r+2} \\ \vdots \\ C_N \end{bmatrix}. \quad (\text{A16})$$

Then by assumption,

$$C'_2[B'_1 : A(\mathbf{D}')B'_1 : \cdots : A(\mathbf{D}')^{n-1}B'_1] = 0. \quad (\text{A17})$$

This is because if (A17) does not hold, then there are $p \in N'_1$ and $q \in N'_2$ such that

$$C_q[B_p : A(\mathbf{D}')B_p : \cdots : A(\mathbf{D}')^{n-1}B_p] \neq 0. \quad (\text{A18})$$

Equation (A18) implies that there is an arc from p to q , i.e., there is a path from i to q in $G(\mathbf{D}')$, contradicting (A10) and (A11). Equation (A17) means that there is a nonsingular matrix T such that

$$\begin{aligned} & \left(\begin{bmatrix} C'_1 \\ C'_2 \end{bmatrix} T^{-1}, TA(\mathbf{D}')T^{-1}, T[B'_1 : B'_2] \right) \\ &= \left(\begin{bmatrix} C'_{11} & C'_{12} \\ 0 & C'_{22} \end{bmatrix}, \begin{bmatrix} A'_{11} & A'_{12} \\ 0 & A'_{22} \end{bmatrix}, \begin{bmatrix} B'_{11} & B'_{12} \\ 0 & B'_{22} \end{bmatrix} \right) \end{aligned} \quad (\text{A19})$$

where the partition of the right side corresponds to that of the left side. From (A14), (A15), (A16), and (A17), we have

$$TB_i = \begin{bmatrix} B_{1i} \\ 0 \end{bmatrix}, \quad i = 1, 2, \dots, r, \quad (\text{A20})$$

$$C_j T^{-1} = [0 : C_{j2}], \quad j = r+1, r+2, \dots, N. \quad (\text{A21})$$

Therefore, for any $\mathbf{D} \in \mathcal{D}$ and any integer p ,

$$\begin{aligned} & C_N \left(A(\mathbf{D}') + \sum_{s=1}^N B_s D_s C_s \right)^{p-1} B_1 \\ &= [0 : C_{N2}] \begin{bmatrix} A'_{11} & A'_{12} \\ 0 & A'_{22} \end{bmatrix}^{p-1} \begin{bmatrix} B'_{11} \\ 0 \end{bmatrix} = 0 \end{aligned} \quad (\text{A22})$$

where A'_{ij} is an appropriate matrix. This means that G° does not have an arc from 1 to N . Hence the property 2) holds. $\square$

APPENDIX B

Proof of Proposition 4:

Proposition 4 will be proved in two steps. First of all, subspaces $\hat{\mathcal{R}}_i(\mathbf{D})$ and $\hat{\mathcal{K}}_i(\mathbf{D})$ are introduced for the i th equivalence class and it is shown that the subspace $\hat{\mathcal{K}}_i$ given by

$$\hat{\mathcal{K}}_i = \hat{\mathcal{R}}_i(\mathbf{D}) \cap \hat{\mathcal{K}}_i^\perp(\mathbf{D}) \quad (\text{B1})$$

is invariant under local output feedback laws. Secondly, it is shown that

$$\tilde{\mathcal{K}}_i = \hat{\mathcal{K}}_i. \quad (\text{B2})$$

Step 1: Assume that node i belongs to N_i , and define subsets N_i^+ and N_i^- as follows:

$$N_i^+ = \{j \in N: i \rightarrow j\}, \quad (\text{B3})$$

$$N_i^- = \{j \in N: j \rightarrow i\}; \quad (\text{B4})$$

then clearly

$$N_i = N_i^+ \cap N_i^- \quad (\text{B5})$$

and N_i^+ and N_i^- are invariant sets under D .

We assume that

$$N_i^+ = \{i_1, i_2, \dots, i_r\}, \quad (\text{B6})$$

$$N_i^- = \{j_1, j_2, \dots, j_s\}, \quad (\text{B7})$$

and construct matrices $\hat{B}_i$ and $\hat{C}_i$, and subspaces $\hat{\mathcal{R}}_i(D)$ and $\hat{\mathcal{K}}_i(D)$ as follows:

$$\hat{B}_i = [B_{i_1} : B_{i_2} : \dots : B_{i_r}], \quad \hat{C}_i = \begin{bmatrix} C_{j_1} \\ C_{j_2} \\ \vdots \\ C_{j_s} \end{bmatrix}, \quad (\text{B8})$$

$$\hat{\mathcal{R}}_i = \text{Im}\{\hat{B}_i\}, \quad (\text{B9})$$

$$\hat{\mathcal{K}}_i = \text{Ker}\{\hat{C}_i\}, \quad (\text{B10})$$

$$\hat{\mathcal{R}}_i(D) = \langle A(D)|\hat{\mathcal{R}}_i \rangle \quad (\text{B11})$$

and

$$\hat{\mathcal{K}}_i(D) = \bigcap_{s=1}^N A(D)^{s-1} \hat{\mathcal{K}}_i \quad (\text{B12})$$

where $\hat{\mathcal{R}}_i(D)$ and $\hat{\mathcal{K}}_i(D)$ are the controllable subspace and the unobservable subspace for the triple $(\hat{C}_i, A(D), \hat{B}_i)$, respectively, and $i = 1, 2, \dots, \tilde{N}$.

Clearly, we have

$$\hat{\mathcal{R}}_i(0) \supset \hat{\mathcal{R}}_i \quad \text{and} \quad \hat{\mathcal{K}}_i(0) \subset \hat{\mathcal{K}}_i \quad (\text{B13})$$

where $\hat{\mathcal{R}}_i$ and $\hat{\mathcal{K}}_i$ are given by (11) and (12).

Lemma A.1: $\hat{\mathcal{R}}_i(D)$ and $\hat{\mathcal{K}}_i(D)$ are invariant under a local output feedback law D .

Proof: By the duality, it is enough to prove Lemma A.1 with respect to $\hat{\mathcal{R}}_i(D)$. Since $s \notin N_i^+$ means

$$C_s [\hat{B}_i : A\hat{B}_i : \dots : A^{n-1}\hat{B}_i] = 0, \quad (\text{B14})$$

we have

$$\begin{aligned} \hat{\mathcal{R}}_i(D) &= \langle A(D)|\hat{\mathcal{R}}_i \rangle \\ &= \left\langle \left(A + \sum_{s \in N_i} B_s D_s C_s \right) | \hat{\mathcal{R}}_i \right\rangle \\ &= \langle A | \hat{\mathcal{R}}_i \rangle. \end{aligned} \quad (\text{B15})$$

Hence Lemma A.1 holds. $\square$

By Lemma A.1, it is clear that

$$\hat{\mathcal{K}}_i = \hat{\mathcal{R}}_i(D) \cap \hat{\mathcal{K}}_i^\perp(D) \quad (\text{B16})$$

is invariant under D .

Step 2: Since

$$\tilde{\mathcal{K}}_i \subset \hat{\mathcal{K}}_i, \quad (\text{B17})$$

it is sufficient to show that

$$\dim(\tilde{\mathcal{K}}_i) = \dim(\hat{\mathcal{K}}_i) \quad (\text{B18})$$

where $\dim(\mathcal{X})$ is the dimension of the space $\mathcal{X}$.

Since $N_i^+ \supset N_i$ and $N_i^- \supset N_i$, N_i^+ and N_i^- can be expressed as follows:

$$N_i^+ = N_i \cup \{i_1, i_2, \dots, i_r\}, \quad (\text{B19})$$

$$N_i^- = N_i \cup \{j_1, j_2, \dots, j_s\}. \quad (\text{B20})$$

Corresponding to (B19) and (B20), we rewrite $\hat{B}_i$ and $\hat{C}_i$ as follows:

$$\hat{B}_i = [\tilde{B}_i : B_{i_r} : \dots : B_{i_r}], \quad \hat{C}_i = \begin{bmatrix} \tilde{C}_i \\ C_{j_r} \\ \vdots \\ C_{j_s} \end{bmatrix}. \quad (\text{B21})$$

Then for any positive p , we have

$$\hat{C}_i A^{p-1} \hat{B}_i = \begin{bmatrix} \tilde{C}_i A^{p-1} \tilde{B}_i & 0 \\ 0 & 0 \end{bmatrix}. \quad (\text{B22})$$

Therefore

$$\begin{aligned} \dim(\hat{\mathcal{K}}_i) &= \text{rank} \begin{bmatrix} \hat{C}_i \hat{B}_i & \dots & \hat{C}_i A^{n-1} \hat{B}_i \\ \vdots & & \vdots \\ \hat{C}_i A^{n-1} \hat{B}_i & \dots & \hat{C}_i A^{2n-2} \hat{B}_i \end{bmatrix} \\ &= \text{rank} \begin{bmatrix} \tilde{C}_i \tilde{B}_i & \dots & \tilde{C}_i A^{n-1} \tilde{B}_i \\ \vdots & & \vdots \\ \tilde{C}_i A^{n-1} \tilde{B}_i & \dots & \tilde{C}_i A^{2n-2} \tilde{B}_i \end{bmatrix} \\ &= \dim(\tilde{\mathcal{K}}_i). \end{aligned} \quad (\text{B23})$$

This completes the proof. $\square$

APPENDIX C

Proof of Proposition 6:

The mathematical induction with respect to $\tilde{N}$ is used. First of all, we will show that the proposition is true for $\tilde{N} = 1$. In this case, system (1) has only one equivalence class, i.e., all the stations are connected strongly. Let us transform the triple $(\tilde{C}_1, A, \tilde{B}_1)$ to the Kalman canonical form with a nonsingular matrix T :

$$\begin{aligned}
& (\tilde{C}_1 T^{-1}, T A T^{-1}, T \tilde{B}_1) \\
& = \begin{bmatrix} C_{11} & C_{12} & 0 & 0 \end{bmatrix}, \\
& \times \begin{bmatrix} A_{11} & A_{12} & 0 & 0 \\ 0 & A_{22} & 0 & 0 \\ A_{31} & A_{32} & A_{33} & A_{34} \\ 0 & A_{42} & 0 & A_{44} \end{bmatrix} \begin{bmatrix} B_{11} \\ 0 \\ B_{31} \\ 0 \end{bmatrix} \\
& \quad \quad \quad (C1)
\end{aligned}$$

where

$$\begin{aligned}
& \left(\begin{bmatrix} C_{11} & C_{12} \end{bmatrix}, \begin{bmatrix} A_{11} & A_{12} \\ 0 & A_{22} \end{bmatrix} \right) \\
& \quad \quad \quad \text{and} \quad \left(\begin{bmatrix} A_{11} & 0 \\ A_{31} & A_{33} \end{bmatrix}, \begin{bmatrix} B_{11} \\ B_{31} \end{bmatrix} \right)
\end{aligned}$$

are an observable pair and a controllable pair, respectively. Partition $\tilde{x}(0) = T x(0)$ according to (C1).

$$\tilde{x}(0) = \begin{bmatrix} x_1(0) \\ x_2(0) \\ x_3(0) \\ x_4(0) \end{bmatrix} \quad (C2)$$

where $x_i(0)$ is an n_i -dimensional vector ($i = 1, 2, 3, 4$). Then no station of system (1) can control $x_2(0)$, $x_3(0)$, and $x_4(0)$ into origin because $x_2(0)$ and $x_4(0)$ are not controllable and $x_3(0)$ and $x_4(0)$ are not observable for them. This means that if 1) of Proposition 6 is true, then the subspace corresponding to $x_2(0)$, $x_3(0)$, and $x_4(0)$ does not exist, and the triple $(\tilde{C}_1, A, \tilde{B}_1)$ is complete.

Conversely, if the triple is complete, then $\tilde{\mathcal{X}}_1$ is equal to the state space and Proposition 5 asserts that the stations of system (1) can reconstruct and move any unknown initial state into the origin in a finite time interval in cooperation with one another. This means that if $\tilde{N} = 1$, then system (1) is controllable under the decentralized information structure if and only if the triple $(\tilde{C}_1, A, \tilde{B}_1)$ is complete.

On the other hand, by the property of the Hankel matrix, we have

$$\text{rank } F_1 = \text{the dimension of } \tilde{\mathcal{X}}_1; \quad (C3)$$

therefore 2) holds if and only if $(\tilde{C}_1, A, \tilde{B}_1)$ is complete.

For 3) it is well known that

$$\tilde{\Lambda} = \bigcap_{\mathfrak{D}_1} \{s \in \mathcal{C} : \det[sI_n - (A + \tilde{B}_1 \tilde{D}_1 \tilde{C}_1)] = 0\} = \emptyset \quad (C4)$$

if and only if the triple is complete. Therefore, the proposition is true for $\tilde{N} = 1$.

Now we will show that if the proposition is true for $\tilde{N} = k$, then it is also true for $\tilde{N} = k + 1$. By Assumption (15)

$$1 \Rightarrow 2 \Rightarrow \cdots \Rightarrow k + 1, \quad (C5)$$

we have

$$\tilde{C}_1 [\tilde{B}_j : A \tilde{B}_j : \cdots : A^{n-1} \tilde{B}_j] = 0 \quad (C6)$$

for $j = 2, 3, \dots, k + 1$. Therefore, there is a nonsingular matrix S such that

$$\begin{aligned}
& \begin{pmatrix} \tilde{C}_1 \\ \tilde{C}_2 \\ \vdots \\ \tilde{C}_{k+1} \end{pmatrix} S^{-1}, S A S^{-1}, \\
& S [\tilde{B}_1 : \tilde{B}_2 : \cdots : \tilde{B}_{k+1}], S x(t) \\
& = \begin{pmatrix} \begin{bmatrix} \bar{C}_{11} & 0 \\ \bar{C}_{21} & \bar{C}_{22} \\ \vdots & \vdots \\ \bar{C}_{k+1,1} & \bar{C}_{k+1,2} \end{bmatrix} \begin{bmatrix} \bar{A}_{11} & 0 \\ \bar{A}_{21} & \bar{A}_{22} \end{bmatrix} \\
& \quad \quad \quad \begin{bmatrix} \bar{B}_{11} & 0 & \cdots & 0 \\ \bar{B}_{21} & \bar{B}_{22} & \cdots & \bar{B}_{2,k+1} \end{bmatrix} \begin{bmatrix} \bar{x}_1(t) \\ \bar{x}_2(t) \end{bmatrix} \end{pmatrix}, \quad (C7)
\end{aligned}$$

where $(\bar{C}_{11}, \bar{A}_{11})$ is an observable pair.

By using (C7) we construct two subsystems from system (1) as follows:

$$\dot{\bar{x}}_1(t) = \bar{A}_{11} \bar{x}_1(t) + \bar{B}_{11} u_1(t), \quad (C8a)$$

$$y_1(t) = \bar{C}_{11} \bar{x}_1(t), \quad (C8b)$$

$$\begin{aligned}
\dot{\bar{x}}_2(t) &= \bar{A}_{22} \bar{x}_2(t) + \bar{B}_{22} u_2(t) \\
&\quad + \cdots + \bar{B}_{2,k+1} u_{k+1}(t), \quad (C9a)
\end{aligned}$$

$$y_j(t) = \bar{C}_{j2} \bar{x}_2(t) \quad (C9b)$$

where $j = 1, 2, \dots, k + 1$. Thus system (C8) has one class and system (C9) has k classes.

Now assume that both systems (C8) and (C9) are controllable under their decentralized information structure, respectively. Then stations of the first class can move $\bar{x}_1(0)$ into the origin in a finite time interval, say $[0, T_1]$. After this, if the other stations move $\bar{x}_2(T_1)$ into the origin in $[T_1, T_2]$, then any unknown initial state $x(0)$ will be moved

into the origin by the time T_2 because the action of stations of the j th class ($j = 2, 3, \dots, k+1$) does not affect $\bar{x}_1(t)$ by (C7).

Conversely, if either system (C8) or system (C9) is not controllable under the decentralized information structure, then system (1) is not so either. This is because the first class stations cannot observe $\bar{x}_2(t)$ and the others cannot control $\bar{x}_1(t)$, i.e., they cannot help the others.

By the assumption that Proposition 6 holds for $\bar{N} = 1, 2, \dots, k$, both systems (C8) and (C9) are controllable under the decentralized information structure, if and only if

$$\sum_{i=1}^{k+1} \text{rank } F_i = \text{rank } F_1 + \sum_{i=2}^{k+1} \text{rank } F_i = n \quad (\text{C10})$$

or

$$\begin{aligned} \tilde{\Lambda} &= \bigcap_D \left\{ s \in \mathbb{C} : \det \left[sI_n - \left(A + \sum_{i=1}^{k+1} B_i D_i C_i \right) \right] = 0 \right\} \\ &= \bigcap_{D_1} \left\{ s \in \mathbb{C} : \det \left[sI_{n_1} - (\bar{A}_{11} + \bar{B}_{11} \bar{D}_1 \bar{C}_{11}) \right] = 0 \right\} \\ &\quad \cup \bigcap_{D_2, \dots, D_{k+1}} \left\{ s \in \mathbb{C} : \det \left[sI_{n_2} - \left(\bar{A}_{22} + \sum_{i=2}^{k+1} \bar{B}_{2i} \bar{D}_i \bar{C}_{i2} \right) \right] = 0 \right\} \\ &= \emptyset \end{aligned} \quad (\text{C11})$$

where n_i is the dimension of $\bar{x}_i(t)$ ($i = 1, 2$). In (C11) we have used the equality

$$\begin{aligned} \det \begin{vmatrix} sI_{n_1} - M_{11} & 0 \\ -M_{21} & sI_{n_2} - M_{22} \end{vmatrix} \\ = \det[sI_{n_1} - M_{11}] \times \det[sI_{n_2} - M_{22}] \end{aligned} \quad (\text{C12})$$

for any matrices M_{ij} .

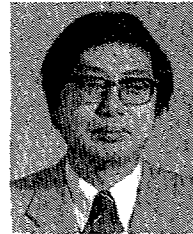
Hence Proposition 6 holds for $\tilde{N} = k+1$. $\square$

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